## NOTES: (UNLESS OTHERWISE SPECIFIED)

1. UFC +5V BUS POWER SWITCHING:

   1.1. JP1 CLOSED, JP2 OPEN, JP3 OPEN
   VBUS POWERS 5V BUS UNTIL EXT PWR (VCC) APPLIED.
   THEN 5V BUS IS POWERED BY VCC.
   (PREFERRED/DEFAULT OPTION)

   1.2. JP1 OPEN, JP2 CLOSED, JP3 OPEN
   5V BUS IS SOLELY POWER BY RAW EXT PWR (VCC).

   1.3. JP1 OPEN, JP2 OPEN, JP3 CLOSED
   5V BUS IS SOLELY POWERED BY RAW USB VBUS.
   CAUTION: FALSE INDICATION OF EXT PWR VIA VCC_SNS
   WILL OCCUR IF EXT PWR CONNECTED.

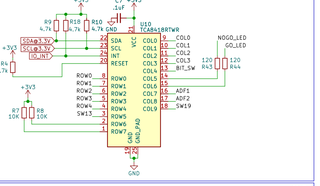
   1.4. JP1 OPEN, JP2 OPEN, JP3 OPEN
   NO 5V POWER IS SUPPLIED TO THE UFC.
   WARNING: UFC WILL NOT FUNCTION IN THIS STATE.

2. ORIENT COMM DISPLAYS (DS7/DS8) SO THE TAB IS TOWARDS THE TOP OF THE PCB.

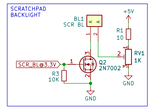
3. ADD PTH RESISTORS TO GO/NO-GO LED CABLES AS REQUIRED TO REDUCE BRIGHTNESS.

4. POPULATE DISPLAYS (BL1, DS1 THRU DS8) WITH TEKCREATIONS TEKDHUFC96 PACKAGE.
(HTTPS://TEKCREATIONS.SPACE)

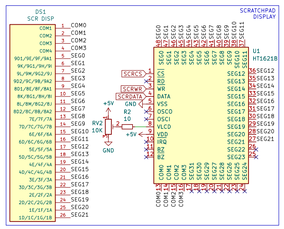
### GPIO EXPANDER



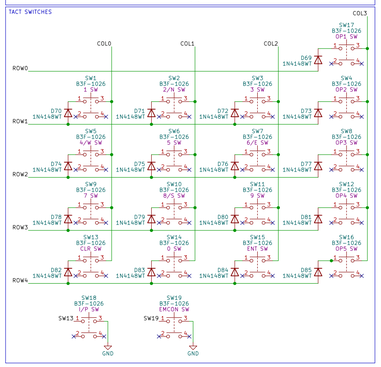

### SCRATCHPAD BACKLIGHT



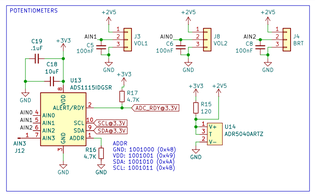

### DS1 SCR DISP / SCRATCHPAD DISPLAY



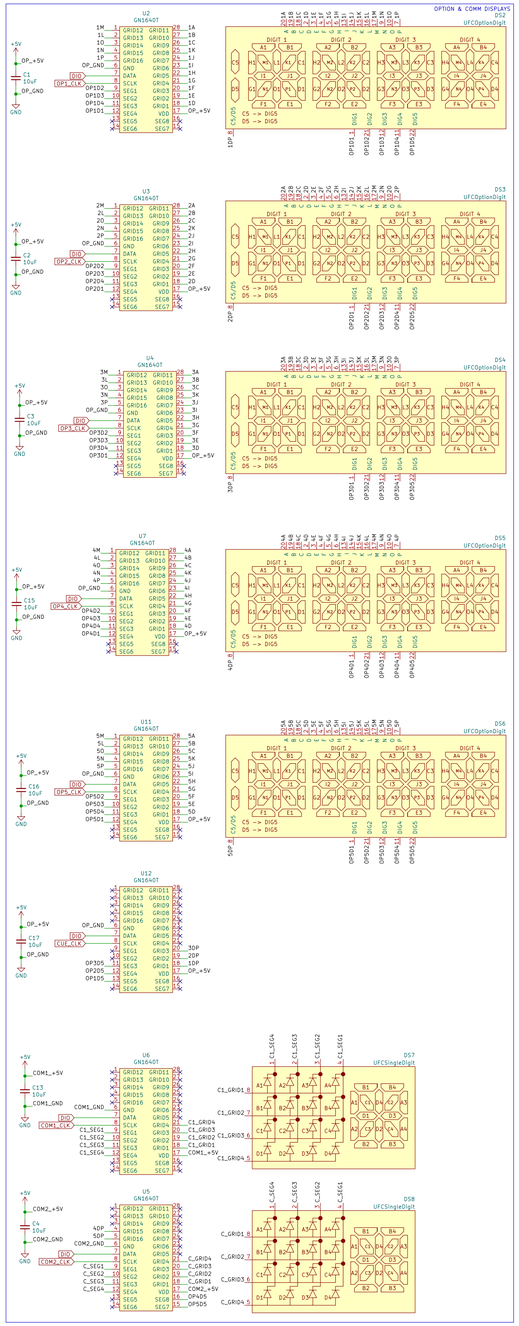

### TACT SWITCHES



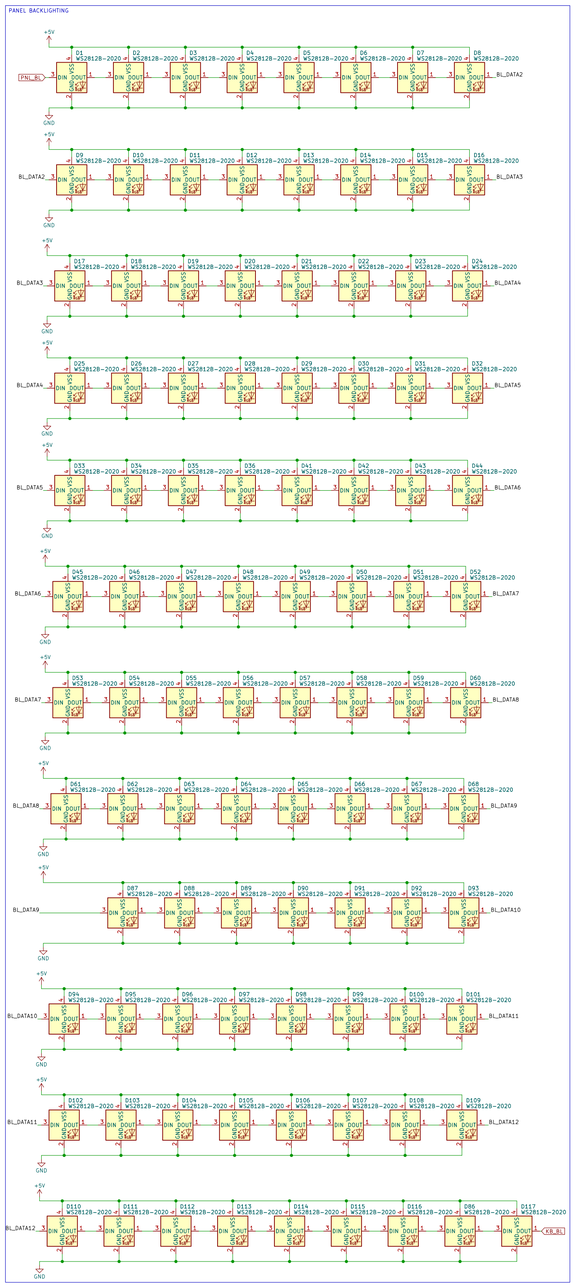

### POTENTIOMETERS



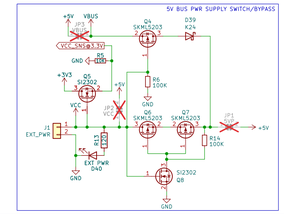

### OPTION & COMM DISPLAYS



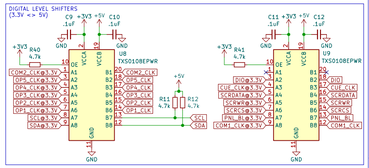

### PANEL BACKLIGHTING



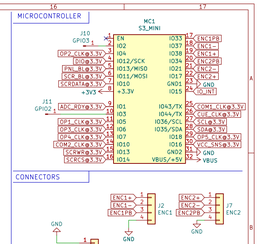

### 5V BUS PWR SUPPLY SWITCH/BYPASS



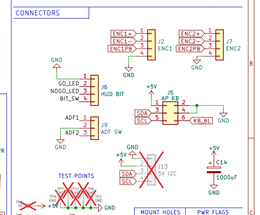

### DIGITAL LEVEL SHIFTERS (3.3V <=> 5V)




### MICROCONTROLLER




### CONNECTORS




OPEN HORNET
www.openhornet.com

| | |
|---|---|
| Sheet: UFC_Main | |
| File: UFC_Main.kicad_sch | |
| **Title: UFC v6 MAIN** | |
| Size: D | Date: 2026-08-30 |
| **Rev: 6.3.1** | |
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